

Deterministic Optimization

Simplex Method in a Nutshell

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

When to Stop and Declare Victory

Simplex Method in a Nutshell

Learning Objectives for this module

- Recognize how to determine stepsize
- Recognize when to stop the algorithm

Move As Much As Possible

- At a BFS, compute reduced costs of all nonbasic variables
 - $\bar{c}_j = c_j - c_B^\top B^{-1} A_j$
 - Pick j such that $\bar{c}_j < 0$
- The corresponding direction $d_B = -B^{-1} A_j$ and $d_N = e_j$
- The new point $x + \theta d$ need to be feasible:
 - $A(x + \theta d) = b$ is automatically satisfied for any $\theta \geq 0$
 - But $x + \theta d \geq 0$ may not be satisfied for some θ
 - We want to make θ as large as possible, because this reduces the objective value as much as possible
 - But $x + \theta d$ should remain nonnegative

Illustration

- Let's say the current BFS is

- $x = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, with basic variables x_1, x_2 and nonbasic variables x_3, x_4, x_5

- The current direction d is to bring x_3 into basis, i.e. make x_3 positive

- $d = \begin{bmatrix} -4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

- At the new point $x + \theta d$: How large θ can be to ensure $x + \theta d \geq 0$?

Illustration cont.

- The new point is

$$\bullet \quad x + \theta d = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \theta \begin{bmatrix} -4 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 - 4\theta \\ 2 - 2\theta \\ \theta \\ 0 \\ 0 \end{bmatrix} \geq \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + \frac{1}{4}d = \begin{bmatrix} 0 \\ 3/2 \\ 1/4 \\ 0 \\ 0 \end{bmatrix}$$

- $1 - 4\theta \geq 0$
- $2 - 2\theta \geq 0$
- Which gives: $\theta \leq 1/4$ and $\theta \leq 1$
- So the largest $\theta = \frac{1}{4}$ is the minimum of the ratios $\frac{x_{B(i)}}{-d_{B(i)}}$ for $d_{B(i)} < 0$

Min-Ratio Test

- In general, to determine the stepsize:
 - We want $x_{B(i)} + \theta d_{B(i)} \geq 0$
 - This can be violated only for $d_{B(i)} < 0$
 - So we should compute:

$$\theta^* = \min_{\{i=1, \dots, m \mid d_{B(i)} < 0\}} \frac{x_{B(i)}}{-d_{B(i)}}$$

- The basic variable $B(i)$ that achieves the min corresponds to $x_{B(i)} + \theta^* d_{B(i)} = 0$, i.e. $x_{B(i)}$ exists the basis.

When To Stop and Declare Victory?

- Either all the reduced costs $\bar{c}_j \geq 0$,
 - Then moving along any direction will not lead to decreasing objective value
 - The algorithm has found the optimal BFS. Stop.
- Or, when $d_B \geq 0$,
 - Then $x_B + \theta d_B \geq 0$ for all $\theta \geq 0$
 - So we can move along d as long as we want and never leave the feasible region.
 - Therefore, the objective value can be driven to $-\infty$.
 - The LP has unbounded optimum. Stop.

Summary

- We learned:
 - How to compute stepsize using min-ratio test
 - When to stop the algorithm